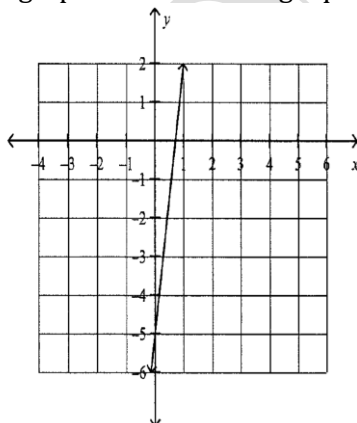
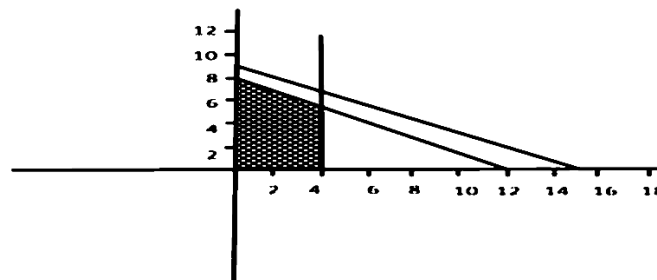


- Q.1** Father's age is 3 times of son's age 5 years ago. After 7 years the father's age will be twice as son age. Calculate the present age of both.
- (a) (7, 31) (b) (5, 35)
(c) (17, 41) (d) (15, 45)
- Q.2** A firm must produce 6,500 units. It can produce in two plants:
 $C_1 = 50,000 + 28x_1$, $C_2 = 8,000 + 25x_2$
 Plant 2 has a capacity of 4,000 units. Find the values of x_1 and x_2 that minimize total cost.
- A) $x_1 = 4,000$, $x_2 = 2,500$ B) $x_1 = 2,500$, $x_2 = 4,000$
 C) $x_1 = 3,250$, $x_2 = 3,250$ D) $x_1 = 6,500$, $x_2 = 0$
- Q.3** A two-digit number is equal to 4 times the sum of the digits and 12 times equals to difference of its digit. If xy represents the required number, solving which of the following equations will lead to required number?
- (a) $xy + 4y = 12$ and $12y + 4x = 1$ (b) $xy + y = 4x$ and $12y + x = 4y$
 (c) $xy - 4x = 4y$ and $xy + 12y = 12x$ (d) $xy + 4y = 12$ and $y + 4x = 12$
- Q.4** A school is planning to organize a sports gala. The cost of organizing one athletics event is Rs. 4,000 and the cost of organizing one non-athletics event is Rs. 8,000. From past experience, the number of athletics events must be 50% more than the number of non-athletics events. If the total budget is Rs. 280,000, how many athletics events should be organized?
- A) 20 B) 25
C) 30 D) 35
- Q.5** Find the slope and y- intercept of the graph of the following equation.



- (a) Slope = $5/7$, y-intercept = 5 (b) Slope = $-5/7$, y-intercept = 5
 (c) Slope = 7, y-intercept = -5 (d) Slope = $5/7$, y-intercept = -5
- Q.6** Umer and Asad invested in a business. The sum of investment of Umer and seven times the Investment of Asad amounts to Rs 18 million. Difference between thrice the investment of Umer and twice the investment of Asad is Rs 8 million. Amounts invested by Umer and Asad is:
- (a) Rs 11 and Rs 1 million respectively
 (b) Rs 7.5 million and Rs 1.5 million respectively
 (c) Rs 4 and Rs 2 million respectively.
 (d) Rs 8 million each
- Q.7** Hamid and Majid invest in a project. If sum of 4 times of Hamid and 5 times of Majid is 32 million and difference of 3 times of Majid and 2 times of Hamid is 6 million find investment of both respectively.
- (a) 3 million & 4 million (b) 3 million & 6 million
 (c) 1 million & 2.5 million (d) None of these
- Q.8** Ten years ago the age of a father was four times his son. Ten years hence the age of the father will be twice that of his son. The present ages of the father and the son are
- (a) (50, 20) (b) (60, 20)
 (c) (55, 25) (d) None of these

- Q.9** Ponting has to pay Rs. 1500,000 in installments. If he pay the first installment of Rs. 80,000 and increase every installment by 1.2 times, then in which year he will pay the amount?
 (a) 8th (b) 9th
 (c) 10th (d) None of these
- Q.10** Shafique and company have developed a new product which would earn revenue of Rs. 90 million during the first year. Thereafter, the revenue would decline by 10% each year. Calculate the revenue that the company would be able to earn over the life of the product.
 (a) 450 million (b) 500 million
 (c) 900 million (d) 1 billion
- Q.11** The infinite G.P. with common ratio of $\frac{1}{2}$ and sum 10 is:
 (a) 5, 2.5, 1.25 (b) 1.25, 2.5, 5....
 (c) 2.5, 1.25, 5 (d) None of these
- Q.12** A person saves Rs. 20,000 in 2022 and increases his annual savings by 10% each year. Find the total savings from 2022 up to and including 2035.
 A) Rs. 490,454 B) Rs. 559,500
 C) Rs. 635,450 D) Rs. 507,728
- Q.13** Sufi Pvt Limited produce two products Oil and ghee which must pass through the same processes in department A and B, having weekly production capacities of 200 hours and 150 hours respectively Oil needs 4 hours in department A and 2 hours in department B. Ghee requires 3 hours and 1 hour respectively. Sufi wants to maximize the profit with given set of inequalities.
 (a) $Z = 700x + 500y$ (b) $Z = 700x + 500y$
 $2x + 3y \leq 200$ $4x + 3y \leq 200$
 $4x + 3y \leq 200$ $2x + y \leq 150 \quad x, y > 0$
 $X, y > 0$ (d) None of these
 (c) $Z = 700x + 500y$
 $4x + 3y \leq 200$
 $2x + y \geq 150$
- Q.14** Which of the following constraints is shown?



- (a) $12x + 8y \leq 24$ (b) $2x + 3y \leq 24$
 $15x + 9y \leq 45$ $3x + 5y \leq 45$
 $x \leq 4$ $x \leq 4$
 $x, y \geq 0$ $x, y \geq 0$
- (c) $8x + 12y \leq 24$ (d) $2x + 3y \leq 24$
 $15x + 9y \leq 45$ $3x + 5y \geq 45$
 $x \leq 4$ $x \leq 4$
 $x, y \geq 0$ $x, y \geq 0$
- Q.15** A factory is planning to buy some machine to produce boxes and has a choice of B-1 or B-9 machines. Rs.9.6 million has been budgeted for the purchase of machines. B-1 machines cost Rs.0.3 million each require 25 hours of maintenance and produce 1,500 units a week. B-9 machines cost Rs.0.6 million each require 10

hour of maintenance and produce 2,000 units a week. Each machine needs 50 square meters of floor area. Floor area of 1,000 square meters and maintenance time of 400 hours are available each week. Since all production can be sold, the factory management wishes to maximize output. Situation of above data in the form of objective function and constraints is:

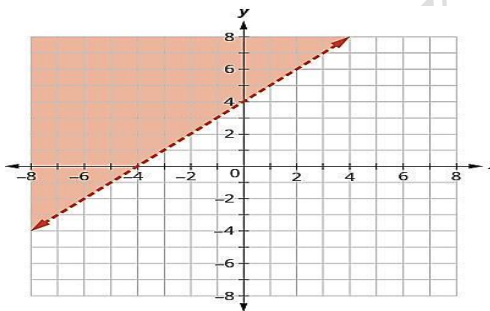
(a) $Z = 1500x + 2000y$
 $0.3x + 0.6y \geq 9.6$
 $25x + 10y \leq 400$
 $50x + 50y \leq 1000$
 $x \geq 0, y \geq 0$

(c) $Z = 1500x + 2000y$
 $0.3x + 0.6y \leq 9.6$
 $25x + 10y \leq 400$
 $50x + 50y \leq 1000$
 $x \geq 0, y \geq 0$

(b) $Z = 1500x + 2000y$
 $0.3x + 0.6y \geq 9.6$
 $25x + 10y \leq 400$
 $50x + 50y \geq 1000$
 $x \geq 0, y \geq 0$

(d) $Z = 1500x + 2000y$
 $0.3x + 0.6y \leq 9.6$
 $25x + 10y \geq 400$
 $50x + 50y \leq 1000$
 $x \geq 0, y \geq 0$

Q.16 Find constraints using the graph



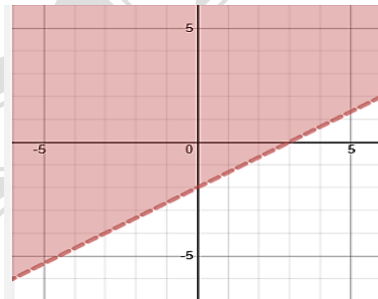
(a) $y > x + 4$

(c) $y \geq x - 4$

(b) $y \leq x + 4$

(d) $y \leq x - 4$

Q.17 Which point below is NOT the part of solution set?



(a) (0,0)

(c) (3,5)

(b) (-10,20)

(d) (0,-2)

Q.18 If $3x+7 \leq 5x-3$, then the inequality holds when x lies in the range.

(a) Less than or equal to 7

(c) Less than or equal to 5

(b) Greater than or equal to 5

(d) Greater than or equal to 2

Q.19 Which of the following statements is/are true?

(i) There is no gap between discrete data.

(ii) Mid-point is calculated by dividing the sum of upper and lower limit by 2 in each class

(a) Both statements are correct

(c) Only statement (i) is correct

(b) Both statements are not correct

(d) Only statement (ii) is correct

Q.20 Which of the following statements is/are correct?

(i) A grouped frequency distribution having class boundaries of discrete data has gaps between the classes.

(ii) In class boundaries, there is no gap between the classes.

- (a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.21** Which of the following is correct?
(a) Data which is collected specifically for ongoing investigation is called raw data
(b) Results of sampling enquiries or a census is called primary data
(c) The data which is relevant to the investigation but was collected previously for some other purpose is called secondary data
(d) All of the above
- Q.22** Construction of a frequency distribution:
(a) Is one of the most common means of summarizing data
(b) Begins by recording the number of times a particular value occurs
(c) Is the basis of construction of a percentage distribution
(d) All of the above
- Q.23** Which of the following statements is/are correct?
(i) A grouped frequency distribution of discrete data has gaps between the classes.
(ii) Discrete data can be converted into continuous data.
(a) Both statements are correct
(b) Both statements are not correct
(c) Only statement (i) is correct
(d) Only statement (ii) is correct
- Q.24** Which of the following is/are correct regarding primary data?
(i) Data collected by interviewing the parents of the students by school management
(ii) Data collected by interviewing the students by school management
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.25** Which of the following statements as regards variance is/are correct?
(i) It can never be smaller than the standard deviation
(ii) It can never be zero.
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.26** Which of the following statements is/are correct?
(i) Range is a measure of dispersion.
(ii) Percentile is a measure of dispersion.
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.27** Consider the following statements about Mean.
(i) It must be one of the values found in the data
(ii) In case of ungroup data, there may be more than one mean.
(a) Both are correct (b) Both are not correct
(c) Only (i) is correct (d) Only (ii) is correct
- Q.28** Which of the following statements is/are correct?
(i) Mean is a measure of central tendency
(ii) Range is a measure of dispersion
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.29** Which of the following statements is/are correct?
(i) Semi-inter Quartile Range is a measure of dispersion.
(ii) Percentile is a measure of dispersion.
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.30** 59, 81, 81, 81, 81, 21, 21, 21, 21, 21, 90, 90, 90, 90. Find mode of the data?
(a) 21 only (b) 21 & 81 only

- (c) 21 & 90 only (d) None of these
- Q.31** Which of the following statements is/are correct?
- (i) Range is a measure of dispersion.
(ii) Semi-inter Quartile Range is a measure of dispersion.
- (a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct

- Q.32** Following data is given, Find weighted mean of both

W	Salary A	Salary B
5	10,000	8,000
3	15,000	10,000
2	20,000	12,000
1	30,000	20,000

- (a) A = 14,000, B = 10,364 (b) A = 15,500, B = 10,363
(c) A = 15,000, B = 10,364 (d) None of these
- Q.33** Following is the data related to number of persons per house in a village town:

No. of persons per	1	2	3	4	5	6	7	8	9	10
No. of houses	25	114	120	90	50	41	20	12	3	2

The mean, median and modal number of persons per house is:

- (a) 3.67, 3, 3 (b) 3.62, 4, 5
(c) 3.67, 3, 5 (d) 3.42, 3, 3
- Q.34** The scores obtained by six students in a set of examination are 80, 40, 50, 72, 45 and 81. These scores are increased by 15%. What will be the effect of these changes on the mean and standard deviation?
- (a) Mean and standard deviation will remain unchanged
(b) Mean will increase by 15% but standard deviation will remain unchanged
(c) Mean and standard deviation both will increase by 15%
(d) Mean will remain unchanged but standard deviation will increase by 15%
- Q.35** Which of the following statements is/are correct?
- (i) Quartile is a measure of dispersion.
(ii) Percentile is a measure of dispersion.
- (a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.36** Average means?
- (a) Summarize the concentration of a set of data
(b) Measures the concentration of a set of data
(c) Measures the scatterness of a set of data
(d) None of these
- Q.37** What is true about mean?
- (a) Best average to describe data (b) It is affected by extreme values
(c) Further algebraic manipulation is possible (d) All of these
- Q.38** Find harmonic mean of 78, 79, 80.
- (a) 78.01 (b) 79.01
(c) 78.99 (d) 78.5
- Q.39** 15, 16, 23, 23, 23, 25, 25, 25, 27, 27, 28, 28, 28, 28, 28, 28, 29, 29, 29, 29
Find mode
- (a) 25 (b) 27
(c) 28 (d) 29
- Q.40** Find upper quartile from the following data: 11, 19, 19, 20, 21, 24, 25, 25, 36
- (a) 20 (b) 25
(c) 19 (d) 24
- Q.41** Data given

Classes	Frequency
46-50	15
51-55	25
56-60	105
61-65	170
66-70	180
71-75	150
76-80	80
81-85	60
86-90	48

Find standard deviation and variance

- (a) 8.17 and 80.02 (b) 9.17 and 84.02
(c) 7.17 and 74.02 (d) 6.17 and 64.02

Q.42 Arman got a rise of 12%, 20% and 18% in 2011, 2012 and 2013 respectively. The average annual increase rate is:

- (a) 16.5% (b) 16.62%
(c) 17.1% (d) 17.33%

Q.43 Over a period of 4 years, Khursheed's salary has increased in the ratios, 1.072, 1.086, 1.069 and 1.098. The average of these ratios and hence the average percent increase is:

- (a) 1.08118 and 8.118% (b) 1.086 and 8.6%
(c) 1.08119 and 8.6% (d) None of these

Q.44 If a regression line passes through the origin, i.e., then the value of a ?

- A) $a = 1$ B) $a = 0$
C) $a = b$ D) $a = -b$

Q.45 Which of the following statement/s is/are correct about class boundary?

- (i) There is no gap between the classes
(ii) Mid-Point is the average of upper-class boundary and lower-class boundary of the same class
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct

Q.46 Which of the following statement/s is/are correct?

- (i) Mean can never be smaller than 1st Quartile
(ii) Mode can never be smaller than 1st Quartile
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct

Q.47 A person invested Rs. 120,000 at simple interest. There is no penalty if he withdraws the amount after 2 years, and he received Rs. 156,000 on withdrawal. Which of the following statements is correct?

Statements:

1. The simple interest rate is 9% p.a.
2. The amount of interest will be Rs. 27,000 if he withdraws after 3 months

- A) Only statement 1 is correct B) Only statement 2 is correct
C) Both statements are correct D) Both statements are not correct

Q.48 Which of the following statement/statements is/are correct?

1. Mean uses every value in the distribution
2. Standard deviation uses every value in the distribution
3. Median is not influenced by extreme values
4. Mean of a discrete data may result in a figure that is not part of the data
(a) 1 (b) 1 and 2
(c) 1, 2 and 4 (d) All are correct

Q.49

Year	Price (Rs.)
20X7	3700
20X8	4500
20X9	4800
20Y0	5000
20Y1	5300

Keeping 20X7 as base year, simple index numbers relative to price for the given period would be:

- (a) 100, 121.62, 129.73, 135.14, 143.24
- (b) 121.62, 100, 93.75, 90, 84.91
- (c) 100, 82.22, 77.08, 74, 69.81
- (d) 82.22, 100, 106.67, 111.11, 117.78

Q.50 Calculate Fisher's Price Index for the year 2011.

Commodities	Price (Rs)		Quantity (kg)	
	2010	2011	2010	2011
Alpha	64	75	270	276
Beta	40	45	124	118
Gamma	18	21	130	121
Eta	58	68	185	267

- (a) 105.57
- (b) 110.44
- (c) 116.57
- (d) None of these

Q.51 Which of the following statement is/are correct?

- (i) In Laspeyre price index, the denominator does not change from year to year.
- (ii) In Paasche's price index, the denominator has to be recalculated every year to take account of the most recent quantities consumed
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

Q.52 Which of the following statements about Laspayer's Price Index is/are correct?

- (i) It tends to overstate inflation.
- (ii) The denominator does not change from year to year.
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

Q.53 Which of the following statements about Laspayer's Price Index is/are correct?

- (i) It tends to overstate inflation.
- (ii) The denominator changes from year to year.
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

Q.54 Chemical Master Company (CMC) produces a special industrial chemical that is a blend of four chemical ingredients. The prices at the beginning and the end of year of each material and quantities required to make one unit of finished product are given below:

Ingredients	Jan 20X2 (Base Period)		Dec 20X5 (Current Period)	
	Price per kg	Kg per unit	Price per kg	Kg per unit
A	0.50	10.00	3.95	11.00
B	8.75	3.00	9.90	2.50
C	0.99	2.00	0.95	3.00
D	4.00	2.00	4.50	5.00

Find Laspeyre's Price index.

- (a) 143%
- (b) 132%
- (c) 194%
- (d) 167%

Q.55 Following Consumer Price index (CPI) has been computed taking 20X8 as base year:

Year	CPI
20X8	104.96
20X9	100
20Y0	116.19
20Y1	115.11
20Y2	132.01

Yearly inflation/Deflation for the above data would be

- (a) (4.98%), 5.83%, 7.75%, 12.80%
- (b) 4.98%, 6.19%, 8.41%, 14.69%
- (c) (4.74%), 16.19%, (0.92)%, 14.68%
- (d) 4.74%, 6.19%, 8.40%, 14.68%

Q.56 Laspeyre price index tends to overstate inflation, this is because

- (a) Both prices and quantities of current year are used for inflation
- (b) Both prices and quantities of base year are used as numerator for inflation
- (c) Consumers reacts to price increases
- (d) All of these

Q.57 Compute the real wages for each of the above years.

Year	Pay (Rs.)	Price Index
2005	12350	110.1
2006	13500	122.3
2007	14800	137.6
2008	16500	160.2

- (a) 11217, 11038, 10766, 10300
- (b) 11217, 11038, 10756, 10330
- (c) 11217, 11138, 10756, 10300
- (d) 11217, 11038, 10756, 10300

Q.58 The price of commodity in different years is given below: compute the Link relative for the following years.

Years	2010	2011	2012	2013
Price	49	53	58	62

- (a) 100, 115.50, 118.40, 112.80
- (b) 100, 92.50, 91.40, 93.55
- (c) 100, 108.16, 109.43, 106.89
- (d) 100, 105.50, 116.50, 114.50

Q.59 Which of the following statement/s about scatter diagram is/are correct?

- (i) It is least important to establish which variable is independent
- (ii) It might indicate a relationship where there is none.
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

Q.60 Which of the following statement/s about scatter diagram is/are correct?

- (i) It leads to correct conclusion even if there are only few data points.
- (ii) It might indicate a relationship where there is none.
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

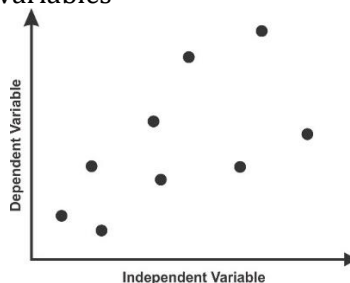
Q.61 Which of the following statement/s about scatter diagram is/are correct?

- (i) It is least important to establish which variable is independent
- (ii) It leads to correct conclusion even if there are only few data points.
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

Q.62 Which of the following statement/s about scatter diagram is/are correct?

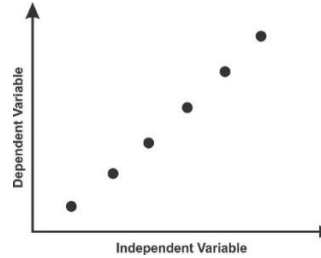
- (i) Shows visual relation between variables.
- (ii) It might indicate a relationship where there is none.
- (a) Both statements are correct
- (b) Both statements are not correct
- (c) Only statement (i) is correct
- (d) Only statement (ii) is correct

- Q.63** Which of the following statement/s as regards to the value of the correlation coefficient is/are correct?
 (i) It is always greater than 0
 (ii) It always lies in the range -1 to +1
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.64** Which of the following statement/s about scatter diagram is/are correct?
 (i) It leads to more accurate results if the collected data is a typical data.
 (ii) It might indicate a relationship where there is none
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.65** Which of the following statement/s is/are correct?
 (i) The coefficient of determination is the square of the correlation coefficient
 (ii) The value of the coefficient of determination must always be in the range 0 to +1.
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.66** Which of the following statement/s is/are correct?
 (i) There is strong relationship between two variables; the points on the scatter diagram would be concentrated around a curve.
 (ii) Linear regression analysis can be used to estimate fixed cost and variable cost per unit from historical total cost and production data.
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.67** Which of the following statement/s is/are correct where Regression is used?
 (i) It is very important in accounting and other business operations
 (ii) It is widely used in economics & business
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.68** Which of the following statement/s is/are correct where Regression is used?
 (i) It tells fixed cost but not variable cost per unit.
 (ii) It shows the extent of relationship between two variables
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.69** Which of the following statement/s is/are correct about correlation coefficient?
 (i) It is always from 0 to +1
 (ii) It is always from 0 to -1
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.70** Find the relationship between two variables



- (a) Strong/perfect positive correlation (b) Moderate positive correlation
 (c) Weak position correlation (d) No correlation

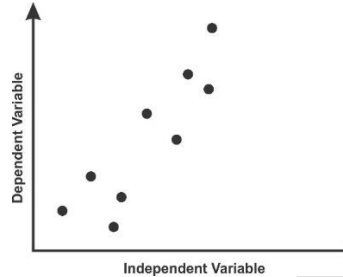
Q.71 Find the relationship between two variables.



- (a) Strong / perfect positive correlation
(c) Weak positive correction

- (b) Moderate positive correlation
(d) No correlation

Q.72 Find the relationship between two variables



- (a) Strong/perfect positive correlation
(c) Weak positive correlation

- (b) Moderate positive correlation
(d) No correlation

Q.73 $n = 10$, $\sum x = 120$, $\sum y = 150$, $\sum x^2 = 1200$, $\sum xy = 900$. Find regression coefficient y on x .

- (a) 3.01
(c) 2.75

- (b) 3.75
(d) 1.75

Q.74 Which of the following correct about scatter diagram?

- (a) Perfect correlation (if all the points lie on the line of regression)
(b) Perfect -ve correlation (if all the points lie on the line of regression sloping downward)
(c) +ve correlation (if scatter diagram shows increasing trend on right side)
(d) All of these

Q.75 Properties of scatter diagram

- (a) Graphical representation of relation b/w independent and dependent variables
(b) It is only used to 2 variables
(c) Independent variable is usually x , dependent variable is usually y
(d) All of these

Q.76 Which of the following statements is/are correct as regards to scatter diagram? (Any Two)

- (a) It is least important to establish which variable is independent before plotting the scatter diagram
(b) It cannot lead to correct conclusions if there are only few data points.
(c) It can show relationship between more than two variables.
(d) It may indicate a relationship where there is none

Q.77 If $r^2 = 0.56$, $b_{xy} = 2$, find b_{yx} ?

- (a) 0.18
(c) 0.28

- (b) 0.38
(d) 0.48

Q.78 An asset costs Rs. 500,000, residual value Rs. 50,000, useful life 5 years (straight-line). Book value at end of year 3 is:

- A) 230,000
C) 320,000

- B) 260,000
D) 350,000

Q.79 If $r = 0.8$ (y on x) means that

- (a) 36% variation in y due to x
(c) 64% variation in x due to y

- (b) 36% variation in x due to y
(d) 64% variation in y due to x

- Q.80** Find the coefficient of correlation between x and y if
Regression line of x on y is: $5x - 4y + 2 = 0$
Regression line of y on x is: $x - 5y + 3 = 0$
- (a) 0.2 (b) 0.3
(c) 0.4 (d) 0.5
- Q.81** What is the primary purpose of the "line of best fit" in linear regression?
- (A) To connect all the data points in a scatter plot.
(B) To summarize the overall trend in the data and make predictions.
(C) To force the data to fit a straight line.
(D) To minimize the correlation between variables.
- Q.82** A person deposits an equal amount at the start of each year for 10 years in an account earning 10% p.a. compounded annually. He then plans to withdraw Rs. 1,000,000 at the end of each year for the next 6 years, with the first withdrawal at the end of the 11th year. What annual deposit should he make?
- A) Rs. 220,000 B) Rs. 248,430
C) Rs. 273,273 D) Rs. 300,000
- Q.83** The members of the selection committee rank eight players for selection as follows: rank correlation of the following data?

Player	A	B	C	D	E	F	G	H
Rank (member 1)	1	2	3	4	5	6	7	8
Rank (member 2)	2	1	5	7	3	6	4	8

The coefficient of rank correlation is:

- (a) 0.67 (b) 0.73
(c) 0.47 (d) 0.37
- Q.84** $b_{xy} = -0.41$ $b_{yx} = -0.71$, $r = ?$
- (a) 0.54 (b) -0.54
(c) 0.35 (d) Not possible
- Q.85** A medical research company has developed the following equation for regression line of y on x (line of best fit) for a particular age group $y = 6.93 + 0.38x$ where x represent height in centimeters and y represents weight in kilogram. Using above equation, we can say that
- (a) For each centimeter increase in height, weight will decrease by 0.38 kilograms
(b) For each centimeter increase in height, weight will increase by 0.38 kilograms
(c) For each centimeter increase in height, weight will increase by 6.55 kilograms
(d) For each centimeter increase in height, weight will decrease by 6.55 kilograms
- Q.86** If $a = -12.57$ and $b = 0.35$ then equation for regression line of y on x (line of best fit) would be:
- (a) $Y = -0.35 + 12.57x$ (b) $Y = 0.35 - 12.57x$
(c) $Y = 12.57 - 0.35x$ (d) $Y = -12.57 + 0.35x$
- Q.87** The data on the following table gives the market share of a product and television advertising expenditure:
- | | | | | | |
|----------------------------|----|----|----|----|----|
| Advertising expenditure(X) | 15 | 17 | 13 | 14 | 16 |
| Market share (Y) | 23 | 25 | 21 | 24 | 26 |
- Estimate market share (Y) when advertising expenditure is Rs. 20
- (a) 27.5 (b) 28.8
(c) 29.8 (d) 30.75
- Q.88** The population of Maldives in 2020 was 418403. If its population is increased at an exponential rate of 2%, what will be the population of Maldives in 2025?
- (a) 462407 (b) 461951
(c) 462406 (d) None of these
- Q.89** If selling price for laptop is Rs.1100 and is marked down by 30% in promotional offers then sale price after discount is?
- (a) Rs.1571 (b) Rs.471
(c) Rs.770 (d) Rs.330

- Q.90** Bazooka invested Rs 500,000 for 5 years after which he received a lump sum amount of Rs 762,150. If he earned 10% interest compounded annually during last 2 years, what rate of interest compounded annually did he earn during the first three years?
- (a) 6% (b) 7%
(c) 8% (d) 9%
- Q.91** Azam & Moazzam invested Rs 10 million each, compounded quarterly for 5 years. After 5 years Azam has 20% more than Moazzam. Azam rate is 10%. Find rate of Moazzam.
- (a) 6.02% (b) 6.12%
(c) 6.27% (d) 6%
- Q.92** Ali borrows Rs.500000 from his friend Kashif for the period of 7 years and Kashif receive Rs. 896793 rupees. If the interest rate was 10% c.a for the last 4 years, find the rate of first 3 years?
- (a) 7% (b) 7.74%
(c) 7.47% (d) 4.77%
- Q.93** Mola Jutt and Noori Nuth both invested same amount for 8 years. If Mola jutt's rate of return is 9% and Noori's rate is 10% compounded annually then how much more percent amount Noori will have after 8 years than Jutt?
- (a) 1% (b) 8%
(c) 15.1% (d) 7.5%
- Q.94** If most of the points in a scatter diagram are clustered around a single point, this indicates:
- (a) No relation (b) Positive relation
(c) Negative relation (d) Linear relation
- Q.95** The profit for product X and Y is Rs.120 per unit and Rs.80 per unit respectively. Find the maximum profit under the given constraints: $4x + 5y \leq 51000$ $7x + 4y \leq 56000$
- A) Rs. 960,000 B) Rs. 1,040,000
C) Rs. 1,120,000 D) Rs. 1,530,000
- Q.96** A person borrowed a loan for 10 years but decided to clear it in 5 annual installments. He paid Rs. 20,000 at the end of the first year, and each subsequent installment was x times the previous installment. If the 5th (last) installment was Rs. 22,090, find the value of x .
- A) 1.015 B) 1.025
C) 1.050 D) 1.105
- Q.97** Sum of thrice of Arif and twice of Ali is 18000 and difference of thrice of Arif and twice of Ali is 15000. How much amount Arif and Ali have?
- (a) 5500, 750 (b) 3000, 750
(c) 4500, 750 (d) 2500, 750
- Q.98** Bhutto Jani made an investment of Rs 1.2 million at 12% compounded quarterly & Shah G made investment of Rs 1.8 million at 10% compounded annually. In how many years Bhutto's investment would exceed Shah's investment?
- (a) 17.68 years (b) 17 years
(c) 18 years (d) 21 years
- Q.99** Bunty and Bubly have invested same amounts in two different investment schemes. Bunty is getting a return of 8% compounded annually whereas Bubly gets a return of 10% compounded annually, the amount of Bubly's interest over a period of six years would exceed the amount of Bunty's interest by
- (a) 11.25% (b) 12%
(c) 31.57% (d) 8%
- Q.100** A person deposited Rs. 600,000 in bank at 9% simple interest for 3 years and 3 months. Calculate the amount of money he received at the end of period:
- (a) 665,500 (b) 550,000
(c) 775,500 (d) 770,500
- Q.101** Mr. Kaali invested Rs 400,000 for ten years after which he received a lump sum amount of Rs 900,000. If he earned 8.50% interest compounded quarterly during last five years which of the following rate compounded quarterly did he earn during first five years.
- (a) 7.89% (b) 8.25%

- (c) 8.39% (d) 8.50%
- Q.102** If Naina and Maina both invested same amount for 6 years, the rate of interest are 8% and 9% respectively. Then how much more percent amount Maina will have after 6 years than Naina?
 (a) 5.6% (b) 8%
 (c) 9.45% (d) 9.02%
- Q.103** Four fair dice are rolled simultaneously. What is the probability that the numbers 1, 2, 3, and 4 all appear (in any order)?
 A) $\frac{1}{54}$ B) $\frac{1}{36}$
 C) $\frac{1}{216}$ D) $\frac{1}{1296}$
- Q.104** Four fair dice are rolled. What is the probability that Die 1 shows 1, Die 2 shows 2, Die 3 shows 3, and Die 4 shows 4?
 A) $\frac{1}{54}$ B) $\frac{1}{216}$
 C) $\frac{1}{1296}$ D) $\frac{1}{24}$
- Q.105** How much would Kamran need to invest now in order to have Rs. 10 million after 10 years, if the annual interest on the investment is 7.2% compounded quarterly?
 (a) Rs. 4,247,239 (b) Rs. 5,855,536
 (c) Rs. 4,898,791 (d) Rs. 5,000,000
- Q.106** If the simple interest on a certain amount for 15 months at $7\frac{1}{2}\%$ per annum exceeds the simple interest on the same amount for 8 months at $12\frac{1}{2}\%$ per annum by Rs.32.50, then the amount is:
 (a) Rs.3,000 (b) Rs.3,060
 (c) Rs.3,120 (d) Rs.3,250
- Q.107** If $\sum x = 58$, $\sum y = 313$, $\sum xy = 3,015$, $\sum x^2 = 594$ and $n = 6$ then equation for regression line of y on x (line of best fit) would be:
 (a) $y = 55.26 - 0.32x$ (b) $y = 0.32 - 55.26x$
 (c) $y = 55.26 + 0.32x$ (d) $x = -0.32 + 55.26y$
- Q.108** The population of a small town increases every year by 2% of the population at the beginning of that year. The number of years by which the total increase of population be 40% is:
 (a) 7 years (b) 10 years
 (c) 17 years (d) None of these
- Q.109** Babar Azam has to repay a loan along with interest, three years from now. The amount payable after three years is Rs. 428,000. The amount of loan presently if interest rate is 8% compounded Semi-annually is:
 (a) Rs. 345,161 (b) Rs. 339,760
 (c) Rs. 338,254 (d) Rs. 300,000
- Q.110** To increase present value, the discount rate should be adjusted:
 (a) Upward (b) Downward
 (c) Not relevant (d) Will depend on time period
- Q.111** ____ is the process of accumulating interest on an investment over time to earn more interest.
 (a) Compounding (b) Discounting
 (c) Promoting (d) All of the above
- Q.112** If future value is multiplied with ____ the result is present value
 (a) Discount factor (b) Compound factor
 (c) Rate of interest (d) Number of years
- Q.113** Which of the following statements is/are true?
 1. Simple interest is calculated on the principal amount and accumulated interest of previous periods
 2. Compound interest is calculated on the principal or original amount of a loan
 3. The growth in compound interest is quite rapid as compared to simple interest
 (a) 1 only (b) 2 only
 (c) 1 and 3 (d) 3 only

- Q.114** Project MCG would provide annual inflows of Rs. 650,000 Rs. 500,000, Rs 1,000,000 and Rs 2,500,000 at the end of year 1 to 4 respectively, whereas project ACG would yield annual inflows of Rs 849,650, Rs 1,166,800 and Rs 2,068,000 from year 1 to 3 respectively. The discount rate at which both projects would have same net present value is:
- (a) 16.31% (b) 17.51%
(c) 18.27% (d) 14.9%
- Q.115** Solution Desk Private Limited intends to invest Rs 4 million into a project which would yield 12,14 and 16 percent for three years respectively. The company would also recover the original investment after 3 years. Company's cost of capital is 10%, NPV of the project would be:
- (a) 385,274 (b) 436,364
(c) 480,841 (d) 1,380,015
- Q.116** A development project costing Rs. 2 million is expected to yield Rs. 300,000, Rs. 400,000, Rs. 1,900,000 at the end of each of the next 3 years respectively. The nearest IRR approximation of the project is:
- (a) 10.18% (b) 10.66%
(c) 11.15% (d) 11.51%
- Q.117** The rate of interest is 8% per annum compounded monthly, the value of perpetuity of Rs 2,500 per month would be:
- (a) Rs 375,000 (b) Rs 187,500
(c) Rs 37,500 (d) Rs 31,250
- Q.118** If the rate of interest is 8% per annum compounded quarterly, the value of perpetuity of Rs 3,500 per quarter would be:
- (a) Rs 131,250 (b) Rs 175,000
(c) Rs 262,500 (d) Rs 525,000
- Q.119** NCHM invested 5 million. Return rate was 10%, 12% and 14% per year for first, second and third year respectively. Find NPV if cost of capital is 10% and initial investment also recovered at the end.
- (a) 0.15 million (b) 0.1656 million
(c) 0.2329 million (d) None of these
- Q.120** If the discount rate is 12%, The present value of Rs. X received at the end of each year for the next five years, is equal to:
- (a) 6x (b) 5x
(c) 3.6x (d) 4.03x
- Q.121** Solution desk invested 9 million. Interest rate was 8%, 12% and 14% per year for first, second and third year respectively. Find NPV if cost of capital is 9%
- (a) 0.15 million (b) 0.1563 million
(c) 0.4922 million (d) None of these
- Q.122** In a class of 30 students (18 boys and 12 girls), a team of 2 boys and 2 girls is to be selected. How many different teams can be formed?
- (a) 12,650 (b) 10098
(c) 15,200 (d) 16,800
- Q.123** What are the qualities of perpetuity?
- (a) Used to find purchase price of share
(b) Used to find value of maintenance fund
(c) Used to find initial deposit required for pension scheme
(d) All of these
- Q.124** A Cement Industry intends to invest Rs. 3 million into a project which would yield 10, 12 and 14 percent for three years respectively. The industry would also recover the original investment after 3 years. If the company's cost of capital is 10% the NPV of the project is:
- (a) Rs 139,745 (b) Rs 46,582
(c) Rs 1,046,582 (d) Rs 3,139,745
- Q.125** An Indoor Sports project costing 4 million is expected to yield Rs. 600000, Rs. 800000 & Rs. 4000000 at the end of each year for next 3 years respectively. The IRR of the project is?
- (a) 12.22% (b) 21.22%

- (c) 22.12% (d) 22.21%
- Q.126** Faisal Javed has invested Rs. 700,000 in an investment scheme. In return, he would receive Rs. 74,587 semi-annually in arrears, for the six years. He would not receive any amount afterwards. Find the nominal and effective rate of return of the scheme.
- (a) 8% and 8.16% (b) 8% and 9.16%
(c) 8% and 10.16% (d) 7% and 7.16%
- Q.127** If discount rate is 10% then the present value of Rs Z payable annually for 4 years is?
- (a) 4.17z (b) 3.17z
(c) 6.25z (d) 5.25z
- Q.128** Which of the following statement is CORRECT?
- (a) Perpetuity is a special kind of annuity that never ends
(b) It is impossible to find the future value of perpetuity
(c) It is impossible to find the present value of perpetuity
(d) Both a and b
- Q.129** Which statement about the Payback Period method is most correct?
- A) It considers the time value of money automatically.
B) It ignores cash flows occurring after the payback point.
C) It always selects projects with the highest NPV.
D) It measures profitability in percentage terms.
- Q.130** A project costs Rs. 550,000. Cash inflows are:
- Year 1: 150,000
Year 2: 170,000
Year 3: 190,000
Year 4: 210,000
What is the payback period?
- A) 2.68 years B) 3.00 years
C) 3.19 years D) 3.50 years
- Q.131** Two companies made profits from in different project:
- | | Year 1 | Year 2 | Year 3 | Year 4 |
|-----------|-----------|---------|---------|---------|
| Company A | 900,000 | 600,000 | 300,000 | 900,000 |
| Company B | 1,200,000 | 800,000 | 400,000 | - |
- Find the rate at which NPV of both companies will be same.
- (a) 29.37% (b) 18.58%
(c) 15.37% (d) 16.33%
- Q.132** Which of the following statement is CORRECT?
- (a) Discounting estimates the present day equivalent of a future cash flow at a specified time in the future at a given rate of interest
(b) Multiplying by a discount factor is the same as multiplying by a compounding factor
(c) The present value of a cash flow is not affected by rate of interest.
(d) Present value fails to appraise large projects with multiple cash flows.
- Q.133** Which of the following is true about perpetuity?
- (a) It has unlimited time series (b) It has no future value
(c) Used to find purchase price of a share (d) All of these
- Q.134** Two companies made profits from investments in different projects:
- | | Year 1 | Year 2 | Year 3 |
|-----------|---------|---------|----------|
| Company A | 600,000 | 900,000 | 850,000 |
| Company B | 600,000 | 750,000 | 1200,000 |
- Find the rate at which NPV of both companies will be same
- (a) 1.47% (b) 1.33%
(c) 54.65% (d) 15.2%
- Q.135** An annuity is a (n):

- (a) Level stream of perpetual cash flows.
 (b) Level stream of cash flows occurring for a fixed period of time.
 (c) Increasing stream of perpetual cash flows.
 (d) Increasing stream of cash flows occurring for a fixed period of time.
 (e) Decreasing stream of cash flows occurring for a fixed period of time.
- Q.136** An annuity stream where the payments occur forever is called a(n):
 (a) annuity due (b) indemnity
 (c) perpetuity (d) ordinary annuity.
- Q.137** Mr. Shafique receives a perpetual stream of Rs.3500 per quarter at the rate of 8% compounded quarterly. Calculate the present value of the stream?
 (a) Rs.43750 (b) Rs.17500
 (c) Rs.175000 (d) None of these
- Q.138** A _____ is a constant cash flow for a given number of periods. Fill in the blank.
 (a) Perpetuity (b) discounting factor
 (c) annuity (d) Net present value
- Q.139** A sinking fund is defined as
 (a) saving of a constant amount for future
 (b) saving of a continuously changing amount for benefit in the future
 (c) lending of a constant amount for specific benefit in future
 (d) continuous cash flow for a long time in future
- Q.140** ECS is deciding for an endowment fund which is likely to generate Rs. 20,000 per year. How much can be invested today when the rate of interest is 9%?
 (a) Rs. 18,348 (b) Rs. 222,222
 (c) Rs. 180,00 (d) Rs. 21,8000
- Q.141** In Lahore 70% of people have UPS and 30% have generators. 20% of people have both, find the probability that a randomly selected house has neither UPS nor generator.
 (a) 20% (b) 10%
 (c) 50% (d) 40%
- Q.142** Probability of occurrence and non-occurrence of an event is expressed as a number of which lies between:
 (a) 0 to 1 inclusively (b) - 1 to +1 inclusively
 (c) - 1 to 0 inclusively (d) 0 to 2 inclusively
- Q.143** Two athletes, A and B participate in a race along with other athletes. If the chance of A winning the race is $\frac{1}{6}$ and that of B winning the same race is $\frac{1}{8}$, what is the chance that athletes other than A & B win the race?
 (a) $\frac{3}{9}$ (b) $\frac{2}{7}$
 (c) $\frac{17}{24}$ (d) None of these
- Q.144** In a factory, the Labour force is comprised of 70 % men and 30% women. If 20% of men have two or more children and 5 % of the women have two or more children, what is the probability that a person having more than one child is a woman?
 (a) 0.215 (b) 0.217
 (c) 0.358 (d) 0.097
- Q.145** Two Dices are rolled, 1 red and 1 blue. What is the probability of rolling a 6 on the red or blue dice?
 (a) $\frac{1}{6}$ (b) $\frac{1}{36}$
 (c) $\frac{11}{36}$ (d) $\frac{2}{6}$
- Q.146** There is 5 red and 7 black Suzuki Cars that are up for sale at Pak Wheels. If 2 cars are sold, what is the probability that both are red?
 (a) 0.6364 (b) 0.1515
 (c) 0.3636 (d) 0.4167
- Q.147** A committee is to be made of 4 members from a total of 13 people of which 5 from Punjab, 2 from Balochistan, 4 from Sindh and 2 from KPK, what is the probability that committee contain no member from Punjab?
 (a) 6.8% (b) 0.7%
 (c) 9.8% (d) None of these

- Q.148** There is 5 red and 7 black Alto are up for sale at OLX. If 2 cars are sold, what is the probability that both are black?
- (a) 0.6364 (b) 0.3182
(c) 0.3636 (d) 0.4167
- Q.149** A wallet contains 65 notes of Rs.1000 and 35 notes of Rs.500. Four notes are selected at random. Find the probability that sum of notes is Rs.3000?
- (a) 0.3156 (b) 0.6844
(c) 0.3105 (d) 0.6895
- Q.150** Two events are mutually exclusive if:
- (a) Their probabilities are less than 1
(b) Their probability sum is equal to 1
(c) They contain every possible outcome of an experiment
(d) They cannot occur at the same time.
- Q.151** A bag contains 6 math books and 12 accounting books. If a student randomly selects two books, then find the probability that both of them are math books or accounting books.
- (a) 0.0423 (b) 0.5556
(c) 0.5000 (d) 0.5294
- Q.152** There are 65 currency notes of Rs 1000 and 35 currency notes of Rs 500 in a safe. If 4 notes are selected randomly find the probability that total amount obtained is at least Rs 4000.
- (a) 0.1727 (b) 0.5246
(c) 0.1427 (d) 0.1313
- Q.153** 50% people use cellular mobile and 40% people use landline and 20% people use both. Find the probability that a person selected at random use neither.
- (a) 20% (b) 30%
(c) 40% (d) 50%
- Q.154** Two dices are rolled what is the probability that at least one 6 appears
- (a) 16.67% (b) 33.33%
(c) 31% (d) 30.55%
- Q.155** A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways 3 balls be drawn from the box, if at least one black ball is to be included in the draw
- (a) 32 (b) 48
(c) 64 (d) 72
- Q.156** A sample of 4 different Color T.V sets is randomly selected from a group containing 57 T.V sets out of which 36 are defective. The probability that all the selected T.V are defective is:
- (a) 0.1400 (b) 0.1491
(c) 0.0181 (d) 0
- Q.157** A committee is to be formed from the stated members, one from Sindh, two from Balochistan, two from Punjab and one from KPK, what is probability that one is from each province while forming a 4-member committee.
- (a) 6.8% (b) 12.7%
(c) 9.7% (d) 26.6%
- Q.158** In how many ways the word "LEGEND" be arranged.
- (a) 250 (b) 720
(c) 360 (d) 150
- Q.159** A school increases its fee per student by 10% each year and the number of students by 5% each year. If the total fee collected in the 10th year is Rs. 6,28,894, find the total fee collected in the 1st year.
- (a) Rs. 1,48,853.01 (b) Rs. 1,60,000.00
(c) Rs. 1,71,925.22 (d) Rs. 1,80,000.00
- Q.160** In how many ways word "BINOMIAL" can be arranged, when vowels come together.
- (a) 120 (b) 40320
(c) 2880 (d) 1440

- Q.161** A and B are two mutually exclusive events and $P(A) = 0.4$ and $P(B) = 0.3$. Find $P(A \cup B)$
- (a) 0.58 (b) 0.7
(c) 0.6 (d) 0.42
- Q.162** If two dice are rolled, what is the probability that either the sum of the two will be seven or at least one of the dice will show the number 5.
- (a) $3/36$ (b) $30/36$
(c) $15/36$ (d) $4/5$
- Q.163** In a bag there are 2 green balls, 3 blue and 2 red balls. If you picked two balls randomly. Find the probability that no one is blue?
- (a) 0.2857 (b) 0.3578
(c) 0.1245 (d) 0.5225
- Q.164** Among 18 Players of Lahore Qalandars, there are 2 wicket keepers and 5 bowlers. In how many ways can a team of 11 members be chosen to include only one wicket keeper and at least three bowlers?
- (a) 12100 (b) 12144
(c) 12200 (d) 14400
- Q.165** If $P(A) = 1/4$, $P(B) = 1/2$, $P(A \cup B) = 5/8$, then $P(A \cap B) = ?$
- (a) $5/7$ (b) $11/21$
(c) $1/8$ (d) $1/3$
- Q.166** Out of 6 eggs in a refrigerator, 2 are rotten. From these 3 eggs are selected at random to make a cake. What is the probability that at least one egg is rotten among 3?
- (a) $2/6$ (b) $4/5$
(c) $2/5$ (d) $2/9$
- Q.167** There are total 57 eggs in a basket out of which 36 are defective. If 4 eggs are selected from the box find the probabilities that only 1 is defective:
- (a) $1/11$ (b) $2/15$
(c) $4/33$ (d) None of these
- Q.168** From manufacturing of 25 bulbs, 3 are defective, if the random sample of 5 bulbs are selected from it. What is the probability that selected sample contain exactly 2 defective bulbs.
- (a) 0.087 (b) 0.258
(c) 0.022 (d) 0.098
- Q.169** In a consignment of 25 auto-batteries, 3 are defective. If a random sample of 5 batteries is selected, then probability of having exactly 2 defective batteries in the sample is:
- (a) 5.3% (b) 3.5%
(c) 12% (d) 8.7%
- Q.170** A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways 3 balls be drawn from the box, if atleast one black ball is to be included in the draw?
- (a) 64 (b) 97
(c) 42 (d) 120
- Q.171** If the probability of an event is 0.01, which of the following statements is correct:
- (a) The event is unlikely to occur
(b) The event is expected to occur about 10% of the time
(c) The event cannot occur
(d) None of the above
- Q.172** A vegetable basket contains 12 peppers, of which 3 are red, 4 are green and 5 are yellow. Three peppers are taken, at random and without replacement, from the basket. Find the probability that the three peppers are all different colors?
- (a) 0.273 (b) 0.727
(c) 0.035 (d) None of these

Q.173

Call metering	1 min.	2 min.	3 min.	4 min.
Call package	Rs. 200 for 200 min.	Rs. 300 for 300 min.	Rs. 400 for 400 min.	----
GPS package	Rs. 200 for 200 mb.	Rs. 300 for 300 mb.	Rs. 400 for 400 mb.	----
SMS Package	Rs. 1 for 1 SMS	Rs. 2 for 2 SMS	Rs. 3 for 3 SMS	----

In how many ways can a customer select a service package?

- (a) 108 (b) 90
(c) 13 (d) 100

Q.174 Eight runners have equal chance of winning the race what is the probability that a person wins the bet if he selects top 3 runners in the correct order?

- (a) 5.61% (b) 0.297%
(c) 0.89% (d) 3.98%

Q.175 A company is offering the following tariff chart:

Call Metering	1 Min.	2 Min.	3 Min.	4 Min.
Call Package	2.00 Rs.	4.00 Rs.	5.00 Rs.	6.00 Rs.
SMS Package	10 Rs.	20 Rs.	----	----
Internet Package	500	1000	1500	200

If a customer wants to choose one of these packages, then find the number of ways available to him.

- (a) 108 (b) 128
(c) 100 (d) 196

Q.176 The price of 1 kg of apples is Rs. 5 more than the price of 1 kg of grapes. If the product of their prices is Rs. 84, find the price of grapes.

- (a) Rs. 6 (b) Rs. 7
(c) Rs. 9 (d) Rs. 10

Q.177 Which statement best describes the Discounted Payback Period?

- A) It ignores time value of money but considers post-payback cash flows.
B) It considers time value of money but still ignores post-payback cash flows.
C) It is always equal to IRR.
D) It always equals the accounting payback.

Q.178 There are five runners in a race, how many different arrangements of three winning positions are possible

- (a) 40 (b) 50
(c) 60 (d) 70

Q.179 A team of 4 people is to be formed from 5 men and 7 women. In how many ways can this be done if there must be more women than men in the team?

- (a) 175 (b) 35
(c) 210 (d) 495

Q.180 In a consignment of 25 auto batteries, 3 are defective. If a random sample of 5 batteries is selected, the probability of having exactly 2 defective batteries is?

- (a) 0.053 (b) 0.035
(c) 0.12 (d) 0.087

Q.181 Food panda receives an average of two orders every four minutes. Assuming an approximate Poisson distribution that five or more orders will be received during a period of eight minutes?

- (a) 0.3712 (b) 0.4679
(c) 0.5321 (d) 0.4679

Q.182 The weights of bags of rice packed on a machine are normally distributed with mean = 5.15 kg and standard deviation 0.05kg. If a bag is picked at random, find the probability that it weighs less than 5 kg.

- (a) 0.9987 (b) 0.5013
(c) 0.4987 (d) 0.0014
- Q.183** The weights of bags of rice packed on a machine are normally distributed with mean = 5.05 kg and standard deviation 0.025kg. If a bag is picked at random, find the probability that it weighs less than 5.06 kg is?
(a) 50.82% (b) 1.64%
(c) 65.54% (d) 47.72%
- Q.184** Which of the following statements as regards Normal distribution is/are correct?
(i) Both tails of the distribution approach and meet the horizontal axis at a finite but high value
(ii) Higher standard deviation leads to a flatter curve
(a) Both statements are correct (b) Both statements are not correct
(c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.185** Online booking system of food channel receives an average of two orders every four minutes. Assuming an approximate Poisson distribution that five or more orders will be received during a period of nine minutes?
(a) 0.3712 (b) 0.4679
(c) 0.5321 (d) 0.3679
- Q.186** There are 5 red and 7 black cars for sale at fast wheels. If 2 cars are sold, what is the probability that first is red & second is black?
(a) 0.6364 (b) 0.1515
(c) 0.2651 (d) 0.7349
- Q.187** The weights of bags of rice packed on a machine are normally distributed with mean = 5.15 kg and standard deviation 0.05kg. If a bag is picked at random, find the probability that it weighs less than 5.05 kg is?
(a) 0.9987 (b) 0.0228
(c) 0.9772 (d) 0.0013
- Q.188** You are given 20 true false questions and required to find the probability of at least 4 correct.
(a) 0.7825 (b) 0.8281
(c) 0.9987 (d) 0.6898
- Q.189** A normal distribution has mean 5.03 and standard deviation of 0.03 then find the probability of values between 5 to 5.06
(a) 0.5858 (b) 0.6826
(c) 0.6256 (d) 0.6101
- Q.190** In a normal distribution mean = 5.05 and SD = 0.02 find the area between 5.00 and 5.06.
(a) 0.4214 (b) 0.5258
(c) 0.6100 (d) 0.6853
- Q.191** Fantastic Five Football Club (FFC) team loses 7% of its matches in every session. The probability that FFC will lose exactly one of its next five matches is?
(a) 7% (b) 26.18%
(c) 0.01% (d) 73.82%
- Q.192** FX food channel receives an average of six orders in every four minutes. Assuming an approximate Poisson distribution that five or more orders will be received during a period of four minutes?
(a) 0.2849 (b) 0.7151
(c) 0.4455 (d) 0.5545
- Q.193** If a student randomly guesses 20 multiple choice questions with four possible choices. The probability that the student would get exactly four right answers is 46.23%.
(a) True (b) False
- Q.194** A telephone operator receives on average 2 calls in 3 minutes. Find the probability of receiving more than or equal to 4 calls in 9 minutes:
(a) 0.8488 (b) 0.4521
(c) 0.2152 (d) 0.3521
- Q.195.** Given the following summary statistics for 6 observations:
 $\sum x = 75, \sum y = 270, \sum xy = 3425, \sum x^2 = 1000$ Find the regression line of y on x.

- A) $y = 35 + 0.80x$ B) $y = 35 - 0.80x$
 C) $y = -35 + 0.80x$ D) $y = -35 - 0.80x$
- Q.196** Sharma brothers limited manufacture a component with a mean weight of 5kg and standard deviation of 1kg. If the weights follow a normal distribution. What is the probability that a component picked up at random will be more than 5kg but less than 6kg
 (a) 0.1587 (b) 0.3413
 (c) 0.8413 (d) 0.2783
- Q.197** Unprepared student go for test. There are 10 true false questions in test find the probability that at least 1 is correct.
 (a) 0.9990 (b) 0.99
 (c) 0.9 (d) 1.0
- Q.198** In a Rice mill the bags of rice has mean weight of 5.05 kg and standard deviation of 0.02 kg. if a bag is selected at random then find the probability that its weight is below 5 kg...
 (a) 0.0062 (b) 0.0072
 (c) 0.0082 (d) 0.0052
- Q.199** Marginal profit is best expressed as:
 A) $MR + MC$ B) $MR - MC$
 C) $TC - TR$ D) TR/TC
- Q.200** During peak hours Wi-tribe Call center receives 4 calls per 30 minutes. What is the probability of getting 3 or more calls in an hour?
 (a) 0.9862 (b) 0.2968
 (c) 0.8296 (d) 0.9826
- Q.201** Ahmad received 4 calls in five minutes on average. What is the probability that more than 3 calls will be received in 10 minutes?
 (a) 0.2365 (b) 0.9576
 (c) 0.1525 (d) 0.0268
- Q.202** A telephone operator receives on average 2 calls in 3 minutes. Find the probability of receiving more than or equal to 4 calls in 9 minutes.
 (a) 0.2365 (b) 0.8488
 (c) 0.1525 (d) 0.0268
- Q.203** The percentage of measurements of a normal random variable fall within the interval $\mu \pm 2\sigma$ is:
 (a) At least 75% (b) 95.44%
 (c) 99.74% (d) None of these
- Q.204** The percentage of measurement of a normal random variable fall within the interval is $\mu \pm 3\sigma$ is:
 (a) At least 88.9% (b) 95.44%
 (c) 99.74% (d) None of these
- Q.205** Selling price = Rs. 80, variable cost = Rs. 50, fixed cost = Rs. 420,000. Find break-even units.
 A) 10,500 B) 12,000
 C) 14,000 D) 16,800
- Q.206** Which of the following statement/s as regards Normal distribution is/are correct?
 (i) Both tails of the distribution approach and meet the horizontal axis at a finite but high value
 (ii) Lower standard deviation leads to a flatter curve
 (a) Both statements are correct (b) Both statements are not correct
 (c) Only statement (i) is correct (d) Only statement (ii) is correct
- Q.207** Two salesmen, A and B, have the following sales records:
 • Salesman A: Mean = 50, SD = 5
 • Salesman B: Mean = 60, SD = 10
 Who is more consistent in performance?
 A. Salesman A B. Salesman
 C. Both are equally consistent D. Cannot be determined

- Q.208** Mr. X has a score of 32 and Mr. Y has a score of 36. The group mean is 35, with standard deviations of 9 (for X) and 7 (for Y). What are the Z-scores of Mr. X and Mr. Y?
- A. $X = +0.33, Y = -0.14$ B. $X = -0.33, Y = +0.14$
C. $X = -0.78, Y = +0.71$ D. $X = +0.78, Y = -0.71$
- Q.209** Adnan invested Rs. 15 million, and Adam invested Rs. 12 million in separate projects. Both projects have the same Net Present Value (NPV) and equal time periods. What can we conclude?
- A. Adnan's discount rate was higher than Adam's
B. Adnan's discount rate was lower than Adam's
C. Both had the same discount rate
D. Adnan's project had a shorter life
- Q.210** If two persons were asked to give their opinions about six items, and their opinions were exactly the same, then the correlation between their responses would be:
- A. 0 B. 1
C. -1 D. 0.5
- Q.211** In an economy, the Laspeyres Price Index for a certain year was calculated as 115. What does this indicate?
- A. Prices have fallen by 15% compared to the base year
B. Prices have risen by 15% compared to the base year
C. Prices have doubled since the base year
D. There is no change in prices
- Q.212** If 20 students are to be seated in 5 rows, each row holding 4 students, and there must be at least one student in the front row, the total number of possible seating arrangements is:
- A. $20!$ B. $5! \times 4!$
C. $20! - 16!$ D. Cannot be determined without order condition
- Q.213** In a simple linear regression equation $Y = a + bX$, 'b' represents:
- A. The dependent variable
B. The intercept on Y-axis
C. The slope or rate of change in Y due to X
D. The error term
- Q.214** There are 5 students, and Ali must always sit in the first seat. In how many ways can the students be seated?
- A. $5!$ B. $4!$
C. $6!$ D. $3!$
- Q.215** In a normal distribution, the mean = median = mode when the curve is:
- A. Skewed right B. Skewed left
C. Perfectly symmetrical D. Bimodal
- Q.216** The mean of a normal distribution is 100 and $\sigma = 10$. What is $P(90 < X < 110)$?
- A. 0.34 B. 0.68
C. 0.95 D. 0.99
- Q.217** If $X = 75, \mu = 70, \sigma = 5$, find Z.
- A. 1.0 B. -1.0
C. 0.5 D. 1.5
- Q.218** The Harmonic Mean (HM) is most suitable for:
- A. Rates and ratios (e.g. speed) B. Incomes and profits
C. Prices of commodities D. Simple frequencies
- Q.219** CPI in 2018 = 148. It increases 10% in 2019 and 2020 and decreases 10% each in 2021 and 2022. Find CPI in 2022.
- A. 160.6 B. 145.
C. 162.8 D. 148.0
- Q.220** Which of the following is a limitation of regression analysis?
- A. Regression line cannot find fixed cost directly
B. Regression works only when correlation is positive

- C. Regression requires equal class intervals
D. Regression cannot predict future values
- Q.221** A company sells 500 units each month. Selling price in 1st month = Rs. 100 per unit
Cost in 1st month = Rs. 90 per unit. Every month, price increases by Rs. 10 and cost increases by Rs. 8. Find the total profit earned from the 13th to 19th month (both included).
A. Rs. 110,000 B. Rs. 120,000
C. Rs. 140,000 D. Rs. 150,000
- Q.222** Which is correct within the relevant range?
A) Fixed cost per unit remains constant
B) Total fixed cost remains constant
C) Total variable cost remains constant
D) Variable cost per unit increases automatically with volume
- Q.223** At 8,000 units total cost = Rs. 180,000. At 5,000 units total cost = Rs. 135,000. Variable cost per unit and fixed cost are:
A) VC=15, FC=60,000 B) VC=10, FC=80,000
C) VC=20, FC=20,000 D) VC=15, FC=45,000
- Q.224** If the fee earned by school in 10th year is 5 million and fee is increased each year by 10% and number of students increased by 5%. What was the fee earned by school in 1st year?
a) 3.07 million b) 1.93 million
c) 1.37 million d) 0.81 million
- Q.225** A person invest Rs. 100,000 at the beginning of each year for 6 years. Deposits made in the first 3 years earn 10% p.a simple interest, while deposits made in last three years earn 12% p.a simple interest. Find the total amount accumulated at the end of 6 years.
a) 764000 b) 182000
c) 822000 d) 130000
- Q.226** A customer first chooses Deal A or Deal B with probability $\frac{1}{2}$ each.
- Deal A (Burger + Soda):
Burgers = {Chicken, Beef, chicken Zinger}
Soda sizes = {1.5L, 2.5L, 500ml}
 - Deal B (Main dish + Soda):
Main dishes = {Chicken Biryani, Beef Karahi}
Soda sizes = {1.5L, 2.5L, 500ml}
- Within a chosen deal, all combinations are equally likely.
- I.** Total number of distinct offers possible in Deal A is:
A) 3 B) 6
C) 9 D) 12
- II.** Total number of distinct offers possible in Deal B is:
A) 3 B) 6
C) 8 D) 9
- III.** Total number of distinct offers possible in the entire menu is:
A) 12 B) 15
C) 18 D) 21
- IV.** Probability that the customer selects a deal with 2.5L soda is:
A) $\frac{1}{2}$ B) $\frac{1}{3}$
C) $\frac{2}{3}$ D) $\frac{1}{6}$
- V.** Probability that the selected deal contains a chicken item is:
A) $\frac{7}{12}$ B) $\frac{1}{2}$
C) $\frac{2}{3}$ D) $\frac{5}{12}$

- VI.** Probability that the selected deal contains a chicken item is:
- | | |
|-------------------|-------------------|
| A) $\frac{7}{12}$ | B) $\frac{1}{2}$ |
| C) $\frac{2}{3}$ | D) $\frac{5}{12}$ |
- VII.** Probability that the customer selects Beef Burger with 1.5L soda is:
- | | |
|-------------------|-------------------|
| A) $\frac{1}{6}$ | B) $\frac{1}{9}$ |
| C) $\frac{1}{18}$ | D) $\frac{1}{12}$ |
- Q.227** Ali has Rs. 11.5 million in a savings account earning 10% p.a., compounded annually. He withdraws Rs. 2 million at the end of each year. For how many years can Ali continue to withdraw the full Rs. 2 million (i.e., how many complete yearly withdrawals are possible before the fund becomes insufficient)?
- | | |
|------------|-------------|
| A) 7 years | B) 8 years |
| C) 9 years | D) 10 years |
- Q.228** A person borrowed money for 10 years but decided to clear the loan in 5 annual installments. He paid Rs. 20,000 at the end of the first year, and then each subsequent installment was x times the previous installment. If the 5th (last) installment was Rs. 212,090, find the value of x .
- | | |
|---------|---------|
| A) 1.65 | B) 1.80 |
| C) 1.90 | D) 2.10 |
- Q.229** A consumer spends 40%, 20%, 30%, and 10% of income on goods A, B, C, and D respectively. Prices of A, B, C, D increase by 15%, 20%, 25%, and 30% respectively. Compute the price index.
- | | |
|----------|----------|
| A) 118.5 | B) 120.5 |
| C) 122.5 | D) 125.0 |
- Q.230** The CPI in 2019 is 148. It increases by 10% in 2020 and 2021 and then decreases by 10% in 2022 and 2023. Compute the CPI for 2022.
- | | |
|----------|----------|
| A) 131.9 | B) 148.0 |
| C) 161.2 | D) 179.1 |
- Q.231** A project costing Rs. 2 million is expected to yield Rs. 300,000, Rs. 400,000, and Rs. 1,900,000 at the end of each of the next three years, respectively. The IRR of the project is:
- | | |
|------------|------------|
| (a) 11.51% | (b) 11.15% |
| (c) 10.66% | (d) 10.18% |
- Q.232** Ali and Asif bought snacks. The total cost (in Rs.) is modeled as:
- 5 pieces of pizza and 2 pieces of bread cost Rs. 8
 - 2 pieces of pizza and 7 pieces of bread cost Rs. 9
- If the price of one pizza piece is p and one bread piece is b , find p and b .
- | | |
|---------------------------|---------------------------|
| A) $p = 1.226, b = 0.935$ | B) $p = 0.935, b = 1.226$ |
| C) $p = 1.000, b = 1.000$ | D) $p = 1.500, b = 0.500$ |
- Q.233** A firm has $MR = 450$ and $MC = 450$ at a particular output level. This point is called:
- | | |
|----------------------|--------------------------|
| A) Break-even point | B) Shut-down point |
| C) Profit-maximizing | D) Maximum revenue point |
- Q.234** A product has selling price Rs. 250 per unit, variable cost Rs. 1,000 per unit, and fixed cost Rs. 400,000. Find Contribution per unit (CPU).
- | | |
|--------------|---------------|
| A) Rs. 1,250 | B) Rs. 750 |
| C) Rs. -750 | D) Rs. -1,150 |
- Q.235** "Variable cost per unit" means:
- | |
|--|
| A) Total fixed cost divided by total units |
| B) Change in total cost when one more unit is produced |
| C) Cost per unit that varies with output |
| D) Total cost minus total revenue |
- Q.236** Which statement is correct?
- | |
|---|
| A) Total revenue is always equal to marginal revenue |
| B) Total revenue is maximum when marginal revenue is zero |

- C) Total revenue is zero when marginal revenue is maximum
D) Marginal revenue is maximum when total revenue is decreasing
- Q.237** A person saves in an arithmetic progression. First month saving is Rs. 10,000, and it increases by Rs. 1,000 each month. Total savings after n months are Rs. 75,000. Find n .
A) 5 B) 6
C) 7 D) 8
- Q.238** Which statement best describes a random sample?
A) It is always free from bias
B) Every member of the population has an equal chance of selection
C) It always produces the most accurate estimate
D) It is selected only from convenient respondents
- Q.239** Which statement correctly distinguishes a histogram from a bar chart?
A) Histogram is used for qualitative data; bars must have gaps
B) Bar chart is used for continuous data; bars must touch
C) Histogram is used for continuous data; adjacent bars touch (no gaps)
D) Both are identical; only names differ
- Q.240** A manager must hold conference calls with 3 different groups selected from 8 groups. The calls are scheduled in sequence, so order matters. How many different schedules are possible?
A) 56 B) 336
C) 6561 D) 8!
- Q.241** Which method of data collection is illustrated when:
(i) School staff interview parents of students
(ii) School management interview students
A) Secondary data B) Primary data
C) Information through correspondents D) Mailed questionnaire
- Q.242** A person deposits an equal amount A at the end of each year for 10 years in a fund paying simple interest. The interest rate is 10% p.a. for the first 5 years and 12% p.a. for the next 5 years. If the total interest earned over 10 years is Rs. 20,000,000, find the annual deposit A .
A) Rs. 1,315,789 B) Rs. 2,500,000
C) Rs. 3,846,154 D) Rs. 4,200,000
- Q.243** David wants to invest for 6 years. Which is better?
(a) 15% simple interest (b) 10% compounded annually
(c) 9% compounded semi-annually (d) 8% compounded monthly
- Q.244** Profit is generated when:
A) Contribution margin exceeds fixed cost
B) Fixed cost increases contribution margin
C) Fixed cost is equal to contribution margin
D) Sales equal variable cost
- Q.245** Alpha deposits money for 10 years. Her first deposit is Rs. 20,000. Each year, her deposit increases by x times. If her 5th-year deposit is Rs. 250,265, find the value of x .
A) 1.80 B) 1.88
C) 1.95 D) 2.10
- Q.246** Median less first quartile ($Q_2 - Q_1$) is 128. Third quartile less median ($Q_3 - Q_2$) is 256. What is interquartile range?
A) 128 B) 256
C) 384 D) 512
- Q.247** A firm is considering a marketing campaign that costs Rs. 10,000. If the campaign is launched. There is a 30% chance the firm will earn Rs. 60,000. There is a 70% chance the firm will earn Rs. 20,000. What is the net expected value of the campaign?
A) Rs. 12,000 B) Rs. 22,000
C) Rs. 32,000 D) Rs. 42,000
- Q.248** Which of the following is/are true?

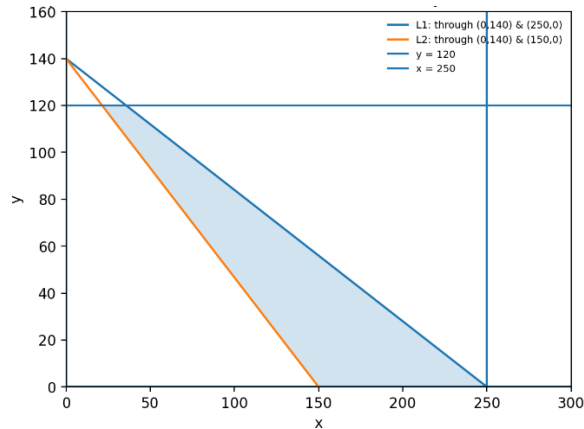
- I. The normal distribution touches the x-axis at a high but finite value.
 II. A higher standard deviation leads to a flatter and wider normal curve.
 A) Only I B) Only II
 C) Both I and II D) Neither I nor II
- Q.249** Bashir invests Rs. 2,000,000 for 1 year at a nominal rate of 12% p.a. Which option gives the highest maturity value?
 A) 12% compounded annually B) 12% compounded semi-annually
 C) 12% compounded quarterly D) 12% compounded monthly
- Q.250** Payback period is best used:
 A) When time value of money is the primary concern
 B) To measure probability of the business
 C) When cash inflows are uncertain and liquidity/recovery speed matters
 D) When a project has a very long life and post-payback inflows dominate
- Q.251** Profit is maximized when:
 A) Marginal cost is zero
 B) Marginal cost is negative
 C) Marginal cost is decreasing
 D) Marginal revenue equals marginal cost and marginal cost is rising (cuts MR from below)
- Q.252** Which statement is correct?
 A) Total angles in a pie chart sum to 180°
 B) Sum of all sector angles in a pie chart equals 360°
 C) Pie chart is most useful when time intervals are concerned
 D) Pie chart is used primarily for continuous class intervals only
- Q.253** Which statement is correct?
 A) Percentile is a measure of dispersion
 B) Range is a measure of dispersion
 C) Both are measures of dispersion
 D) Neither is a measure of dispersion
- Q.254** Team A: mean score = 80, SD = 12
 Team B: mean score = 60, SD = 8
 Which team is more consistent?
 A) Team A B) Team B
 C) Both equally consistent D) Cannot be determined
- Q.255** Which of the following is most closely related to discrete data?
 A) Weight of employees B) Height of employees
 C) Number of children of employees D) Age of employees
- Q.256** An investor wants to receive Rs. 30,000 at the end of each year from Year 4 to Year 10. The interest rate is 10% p.a. What sum should be invested today to fund these withdrawals?
 A) Rs. 99,747 B) Rs. 109,730
 C) Rs. 120,000 D) Rs. 146,040
- Q.257** A machine produces 200 million units in its first year. Due to wear and tear, its annual production declines by 5% each year. The selling price is Rs. 240 per unit. Estimate the total revenue the machine could earn over its life
 A) Rs. 9.6 billion B) Rs. 96 billion
 C) Rs. 960 billion D) Rs. 9,600 billion
- Q.258** How many different arrangements can be made from the letters of the word "BINOMIAL" if all vowels are together?
 A) 720 B) 1,200
 C) 1,440 D) 2,880
- Q.259** Given the values: 98,000; 43,000; 67,000; 43,000, find the mean and mode.
 A) Mean = 62,750; Mode = 43,000 B) Mean = 63,500; Mode = 67,000
 C) Mean = 62,750; Mode = 67,000 D) Mean = 63,500; Mode = 43,000

- A) 8.20% B) 9.80%
C) 10.50% D) 12.00%
- Q.277** A student saves Rs. 1,000 in the first month and then saves Rs. 500 more each month than the previous month. After how many months will the total savings become Rs. 45,000?
A) 10 months B) 11 months
C) 12 months D) 15 months
- Q.278** A furniture item is purchased on instalments. The buyer pays a down payment of Rs. 50,000 today and then pays Rs. 20,000 at the end of each year for 5 years. If the interest rate is 10% p.a. compounded annually, what is the cash price (cost) of the furniture today?
A) Rs. 125,816 B) Rs. 141,654
C) Rs. 150,000 D) Rs. 160,000
- Q.279** Which of the following graphs is most appropriate for a grouped frequency distribution?
A) Pie chart B) Bar chart
C) Histogram D) Stem-and-leaf display
- Q.280** Which of the following statements about regression analysis is correct?
A) Regression proves that one variable causes the other
B) Regression shows the direction and degree of relationship between variables without prediction
C) Regression provides an equation to estimate/predict the value of one variable from the other, but does not prove causation
D) Regression can be applied only when the correlation coefficient is exactly 1
- Q.281** An officer meets on average 16 people per day. Assuming the number of people he meets per day follows a Poisson distribution, what is the probability that on a given day he meets more than 20 people?
A) 0.0682 B) 0.1318
C) 0.2640 D) 0.8682
- Q.282** Which of the following is a key limitation of the payback period method?
A) It considers the time value of money in all cases
B) It ignores cash flows occurring after the payback period
C) It is based on discounted cash flows only
D) It is suitable only for projects with long lives
- Q.283** A company is evaluating a decision whose net expected value (EV) is negative. Which statement is most appropriate?
A) The company will always make a loss if it takes the decision
B) On average, the company is expected to incur a loss if it repeatedly takes similar decisions
C) A negative expected value guarantees a loss in every possible outcome
D) The decision should always be accepted because probabilities may change
- Q.284** A student states: "When marginal cost (MC) = 0, total revenue (TR) is maximum." Which of the following statements is correct?
A) Total revenue is maximum when MC = 0
B) Total revenue is maximum when MR = 0
C) Total revenue is maximum when MC = MR
D) Total revenue is maximum when MR is maximum
- Q.285** Which of the following statements about a random sample is most appropriate?
A) It is suitable only when all population groups are unequal in size
B) It guarantees zero bias in all circumstances
C) It requires a complete sampling frame and may be cumbersome (time-consuming/costly) for large or scattered populations
D) It means selecting respondents who are easiest to reach
- Q.286** Which of the following statements about population is true?
1. It consists of the group of people/items under investigation.
2. Population refers only to human beings, not to objects or events.
A) Only statement 1 is true B) Only statement 2 is true
C) Both statements are true D) Both statements are false

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- A graph of the linear function $y = -x + 140$. The x-axis ranges from 0 to 300, and the y-axis ranges from 0 to 160. The line passes through the points $(0, 140)$ and $(140, 0)$. The area under the line is shaded orange.

(a) $140x + 250y \geq 35,000; x \geq 0; y \geq 0$ (b) $140x + 250y \leq 35,000; x \geq 0; y \geq 0$
(c) $250x + 140y \leq 35,000; x \geq 0; y \geq 0$ (d) $140x + 250y \leq 25,000; x \geq 0; y \geq 0$

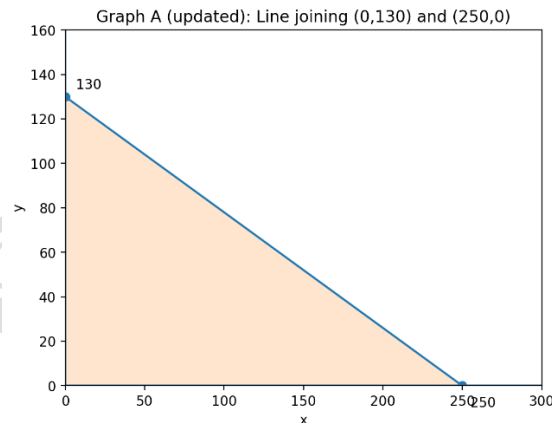
Q.294



Which set of inequalities represents the shaded feasible region?

- (a) $x \geq 0, y \geq 0, x \leq 250, y \leq 120, 140x + 250y \leq 35,000, 140x + 150y \geq 21,000$
 (b) $x \geq 0, y \geq 0, x \leq 250, y \leq 120, 140x + 250y \geq 35,000, 140x + 150y \geq 21,000$
 (c) $x \geq 0, y \geq 0, x \leq 250, y \leq 120, 140x + 250y \leq 35,000, 140x + 150y \leq 21,000$
 (d) $x \geq 0, y \geq 0, x \leq 250, y \leq 120, 250x + 140y \leq 35,000, 140x + 150y \geq 21,000$

Q.295



Which set of inequalities represents the feasible region?

- (a) $130x + 250y \geq 32,500; x \geq 0; y \geq 0$ (b) $130x + 250y \leq 32,500; x \geq 0; y \geq 0$
 (c) $250x + 130y \leq 32,500; x \geq 0; y \geq 0$ (d) $130x + 250y \leq 35,000; x \geq 0; y \geq 0$

کامیابی کا تعویذ

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	B	C	C	C	C	A	A	B	C	A	B	B	B	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	D	B	D	D	D	D	A	A	B	C	B	A	C	A
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A	C	A	C	B	B	D	C	C	B	B	B	A	B	A
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
B	D	D	A	C	A	A	C	C	C	C	D	C	D	D
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
B	A	D	A	A	A	A	B	B	C	A	B	B	D	D
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
B,D	C	A	D	C	B	B	A	B	B	D	B	A	C	C
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
C	A	D	A	B	B	A	A	C	C	A	A	A	C	C
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
C	A	C	C	B	A	A	D	B	A	B	A	B	C	C
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
C	B	D	A	A	A	B	D	B	C	B	A	D	B	B
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
C	C	C	A	B	A	A	C	D	C	B	C	B	A	D
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
D	A	B	D	C	B	D	C	C	D	B	C	A	B	C
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
B	C	A	D	A	A	A	A	B	B	B	B	C	C	D
181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
A	D	C	D	B	C	B	C	B	D	B	B	B	A	A
196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
B	A	A	B	A	B	B	B	C	C	B	A	B	B	B
211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
B	A	C	B	C	B	A	A	B	A	C	B	A	C	C
226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
C,B,B,B A,A,C	B	B	B	C	C	A	C	C	C	B	B	B	C	B
241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
B	C	A	A	B	C	B	B	D	C	D	B	B	B	C
256	257	258	259	260	261	262	263	264	265	266	267	268	269	270
B	C	C	A	C	B	C	A	B	A	A	C	B	D	C
271	272	273	274	275	276	277	278	279	280	281	282	283	284	285
C	B	A	C	A	B	C	A	C	C	B	B	B	B	C
286	287	288	289	290	291	292	293	294	295					
A	B	B	C	C	A	C	B	A	B					